Sri Lanka Institute of Information Technology



SE2030

Software Engineering

2026-Y2-S1-MLB-B3G1-08

Web-based Medicine/Drugs Ordering System

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B.Sc. (Hons) in Information Technology

1. User Stories

1. Customer / Patient (Chronic & Regular Care)

- As a Customer, I want to securely upload digital images or PDFs of my doctor's prescription, so that the licensed pharmacy team can verify and dispense my required medications.
- As a Customer, I want to set up automated monthly refill schedules for chronic illness medications (e.g., diabetes, hypertension), so that I never run out of vital daily prescriptions.
- As a Customer, I want to view the real-time fulfillment status of my order across each stage, so that I have complete transparency regarding dispensing and delivery schedules.
- As a Customer, I want to receive an OTP via SMS and email when my package is dispatched, so that I can securely verify the handover with the courier upon arrival.

2. Licensed Pharmacist (Clinical Verification & Oversight)

- As a Pharmacist, I want to access a dedicated clinical verification portal with digital image inspection tools, so that I can validate doctor credentials, dosage accuracy, and issue dates.
- As a Pharmacist, I want to approve, reject, or request customer clarifications on submitted prescriptions with mandatory audit reason logging, so that strict clinical compliance is enforced.
- As a Pharmacist, I want to tag temperature-sensitive medications with cold-chain packaging instructions, so that pharmacy staff and couriers handle them under strict 2°C–8°C protocols.
- As a Pharmacist, I want to review and clinically renew pending chronic refill requests before automatic order dispatch, so that long-term patient medication safety is guaranteed.

3. Pharmacy Assistant & Inventory Clerk

- As a Pharmacy Assistant, I want the system to allocate medicines using First-Expired-First-Out (FEFO) automated batch selection, so that near-expiry inventory is dispensed first to minimize wastage.
- As a Pharmacy Assistant, I want to receive automated alerts when stock levels fall below critical reorder thresholds, so that inventory replenishments can be ordered proactively.
- As a Pharmacy Assistant, I want the system to automatically quarantine and lock expired batches from active dispensing, so that unsafe medications cannot be accidentally allocated to orders.
- As a Pharmacy Assistant, I want to scan and log batch serial numbers during order packaging, so that every dispensed item is fully traceable to its manufacturer batch.

4. Delivery Courier & Logistics Coordinator

- As a Delivery Courier, I want to view optimized delivery routes and geofenced customer location maps on my mobile interface, so that I can complete dispatches efficiently.
- As a Delivery Courier, I want to input the customer's secure delivery OTP and verify recipient identity, so that controlled pharmaceuticals are handed over only to authorized recipients.
- As a Delivery Courier, I want to log cold-chain carrier box temperatures at dispatch and handover, so that temperature compliance records are maintained throughout transit.
- As a Delivery Coordinator, I want to monitor active delivery fleets and receive automated delay or breach alerts in real time, so that logistics exceptions can be resolved immediately.

5. System Administrator

- As a System Administrator, I want to configure Role-Based Access Control (RBAC) permissions across Customer, Pharmacist, Assistant, and Courier accounts, so that users only access authorized modules.
- As a System Administrator, I want to inspect immutable system audit logs for all clinical approvals, inventory adjustments, and status changes, so that regulatory compliance is strictly maintained.
- As a System Administrator, I want to monitor multi-channel notification gateway health (Email, SMS, WhatsApp), so that transactional messaging runs without interruption.
- As a System Administrator, I want to manage system backup routines and geofencing delivery boundaries, so that operational continuity and service coverage are maintained.

6. Operations & Finance Manager

- As an Operations Manager, I want to view operational dashboards showing prescription approval times and courier transit durations, so that workflow bottlenecks can be eliminated.
- As a Finance Manager, I want to track revenue, payment reconciliation, and pending COD collections, so that accurate daily financial audits are maintained.
- As an Operations Manager, I want to generate stock turnover and expiry wastage reports across all pharmaceutical categories, so that procurement efficiency is maximized.
- As a Finance Manager, I want to process authorized order cancellations and issue partial refunds through integrated payment pathways, so that customer disputes are resolved fairly.

2. Product Backlog

ID	User Story	Priority
PBI - 01	As a Customer, I want to securely upload digital images or PDFs of my doctor's prescription, so that the licensed pharmacy team can verify and dispense my required medications.	**High**
PBI - 02	As a Pharmacist, I want to access a dedicated clinical verification portal with digital image inspection tools, so that I can validate doctor credentials, dosage accuracy, and issue dates.	**High**
PBI - 03	As a Pharmacist, I want to approve, reject, or request customer clarifications on submitted prescriptions with mandatory audit reason logging, so that strict clinical compliance is enforced.	**High**
PBI - 04	As a Customer, I want to view the real-time fulfillment status of my order across each stage, so that I have complete transparency regarding dispensing and delivery schedules.	**High**
PBI - 05	As a System Administrator, I want to configure Role-Based Access Control (RBAC) permissions across Customer, Pharmacist, Assistant, and Courier accounts, so that users only access authorized modules.	**High**
PBI - 06	As a Pharmacy Assistant, I want the system to allocate medicines using First-Expired-First-Out (FEFO) automated batch selection, so that near-expiry inventory is dispensed first.	**High**
PBI - 07	As a Pharmacy Assistant, I want the system to automatically quarantine and lock expired batches from active dispensing, so that unsafe medications cannot be accidentally allocated to orders.	**High**
PBI - 08	As a Pharmacy Assistant, I want to scan and log batch serial numbers during order packaging, so that every dispensed item is fully traceable to its manufacturer batch.	**High**
PBI - 09	As a Pharmacist, I want to tag temperature-sensitive medications with cold-chain packaging instructions, so that pharmacy staff and couriers handle them under strict 2°C–8°C protocols.	**High**
PBI - 10	As a Delivery Courier, I want to input the customer's secure delivery OTP and verify recipient identity, so that controlled pharmaceuticals are handed over only to authorized recipients.	**High**

PBI - 11	As a Delivery Courier, I want to view optimized delivery routes and geofenced customer location maps on my mobile interface, so that I can complete dispatches efficiently.	**High**
PBI - 12	As a System Administrator, I want to inspect immutable system audit logs for all clinical approvals, inventory adjustments, and status changes, so that regulatory compliance is strictly maintained.	**High**
PBI - 13	As a Customer, I want to set up automated monthly refill schedules for chronic illness medications (e.g., diabetes, hypertension), so that I never run out of vital daily prescriptions.	**Medium**
PBI - 14	As a Pharmacist, I want to review and clinically renew pending chronic refill requests before automatic order dispatch, so that long-term patient medication safety is guaranteed.	**Medium**
PBI - 15	As a Delivery Courier, I want to log cold-chain carrier box temperatures at dispatch and handover, so that temperature compliance records are maintained throughout transit.	**Medium**
PBI - 16	As a Delivery Coordinator, I want to monitor active delivery fleets and receive automated delay or breach alerts in real time, so that logistics exceptions can be resolved immediately.	**Medium**
PBI - 17	As a Pharmacy Assistant, I want to receive automated alerts when stock levels fall below critical reorder thresholds, so that inventory replenishments can be ordered proactively.	**Medium**
PBI - 18	As a Finance Manager, I want to process authorized order cancellations and issue partial refunds through integrated payment pathways, so that customer disputes are resolved fairly.	**Medium**
PBI - 19	As a Customer, I want to receive an OTP via SMS and email when my package is dispatched, so that I can securely verify the handover with the courier upon arrival.	**Low**
PBI - 20	As a Operations Manager, I want to view operational dashboards showing prescription approval times and courier transit durations, so that workflow bottlenecks can be eliminated.	**Low**
PBI - 21	As a Finance Manager, I want to track revenue, payment reconciliation, and pending COD collections, so that accurate daily financial audits are maintained.	**Low**

PBI - 22	As a Operations Manager, I want to generate stock turnover and expiry wastage reports across all pharmaceutical categories, so that procurement efficiency is maximized.	**Low**
PBI - 23	As a System Administrator, I want to monitor multi-channel notification gateway health (Email, SMS, WhatsApp), so that transactional messaging runs without interruption.	**Low**
PBI - 24	As a System Administrator, I want to manage system backup routines and geofencing delivery boundaries, so that operational continuity and service coverage are maintained.	**Low**

3. Sprint Plan

1st Sprint – Week 4 to 7

Sprint ID	User Story	Priority
SP1 - 01	As a Customer, I want to securely upload digital images or PDFs of my doctor's prescription, so that the licensed pharmacy team can verify and dispense my required medications.	**High**
SP1 - 02	As a Pharmacist, I want to access a dedicated clinical verification portal with digital image inspection tools, so that I can validate doctor credentials, dosage accuracy, and issue dates.	**High**
SP1 - 03	As a Pharmacist, I want to approve, reject, or request customer clarifications on submitted prescriptions with mandatory audit reason logging, so that strict clinical compliance is enforced.	**High**
SP1 - 04	As a Customer, I want to view the real-time fulfillment status of my order across each stage, so that I have complete transparency regarding dispensing and delivery schedules.	**High**
SP1 - 05	As a System Administrator, I want to configure Role-Based Access Control (RBAC) permissions across Customer, Pharmacist, Assistant, and Courier accounts, so that users only access authorized modules.	**High**
SP1 - 06	As a Pharmacy Assistant, I want the system to allocate medicines using First-Expired-First-Out (FEFO) automated batch selection, so that near-expiry inventory is dispensed first.	**High**

Sprint Backlog – 1st Sprint

1. Prescription Upload Pipeline

Task ID	Task	Est. Hours
T - 01.1	Design responsive drag-and-drop prescription upload UI with client-side preview	4
T - 01.2	Develop secure multipart file upload API with image/PDF validation	4
T - 01.3	Implement SHA-256 cryptographic file hashing to prevent duplicate submissions	3

T - 01.4	Configure encrypted cloud blob storage for prescription documents	3
T - 01.5	Execute integration and upload boundary stress tests	2

2. Clinical Verification Portal

Task ID	Task	Est. Hours
T - 02.1	Design high-resolution prescription inspection UI with pan/zoom utilities	4
T - 02.2	Develop backend API to fetch real-time pending prescription queue	3
T - 02.3	Implement side-by-side patient clinical history and past orders panel	3
T - 02.4	Add doctor SLMC registration number verification checker	2
T - 02.5	Test prescription rendering and inspection interface across browsers	2

3. Prescription Approval & Rejection Logic

Task ID	Task	Est. Hours
T - 03.1	Build interactive approval/rejection modal with structured clinical reason dropdowns	3
T - 03.2	Develop backend state transition API for approved/rejected/clarification status	4
T - 03.3	Implement automated email/SMS alert trigger on prescription status change	3
T - 03.4	Log all pharmacist verification actions into immutable audit database table	2
T - 03.5	Test clinical verification approval and rejection edge cases	2

4. Real-Time Order Tracking Workflow

Task ID	Task	Est. Hours
T - 04.1	Design visual stepper progress UI for multi-stage order tracking	4
T - 04.2	Implement backend state-machine workflow engine (Placed -> Verified -> Packed -> Out)	5
T - 04.3	Establish WebSocket endpoint for real-time order status broadcasting	4

T - 04.4	Display estimated preparation and dispatch timestamps dynamically	2
T - 04.5	Conduct end-to-end order workflow testing from placement to fulfillment	3

5. Role-Based Access Control (RBAC) & Auth

Task ID	Task	Est. Hours
T - 05.1	Define comprehensive permission matrix for all 6 user roles in Clerk / Spring Security	4
T - 05.2	Implement JWT token generation, role claims validation, and route guards	4
T - 05.3	Build administrative user management UI for role assignment and revoking	3
T - 05.4	Implement account brute-force protection and lockout mechanism	2
T - 05.5	Perform security penetration test on RBAC endpoint authorization barriers	3

6. FEFO Batch Allocation Algorithm

Task ID	Task	Est. Hours
T - 06.1	Design batch registration interface (Batch No, Manufacture Date, Expiry Date, Quantity)	4
T - 06.2	Develop FEFO inventory allocation algorithm sorting active batches by earliest expiry	5
T - 06.3	Implement database transaction locks during batch stock deduction	3
T - 06.4	Display allocated batch numbers and shelf locations on pharmacy packing slips	3
T - 06.5	Execute algorithmic unit tests simulating multiple concurrent order batch deductions	3

2nd Sprint – Week 8 to 10

Sprint ID	User Story	Priority
SP2 - 01	As a Pharmacy Assistant, I want the system to automatically quarantine and lock expired batches from active dispensing, so that unsafe medications cannot be accidentally allocated to orders.	**High**
SP2 - 02	As a Pharmacy Assistant, I want to scan and log batch serial numbers during order packaging, so that every dispensed item is fully traceable to its manufacturer batch.	**High**
SP2 - 03	As a Pharmacist, I want to tag temperature-sensitive medications with cold-chain packaging instructions, so that pharmacy staff and couriers handle them under strict 2°C–8°C protocols.	**High**
SP2 - 04	As a Delivery Courier, I want to input the customer's secure delivery OTP and verify recipient identity, so that controlled pharmaceuticals are handed over only to authorized recipients.	**High**
SP2 - 05	As a Delivery Courier, I want to view optimized delivery routes and geofenced customer location maps on my mobile interface, so that I can complete dispatches efficiently.	**High**
SP2 - 06	As a System Administrator, I want to inspect immutable system audit logs for all clinical approvals, inventory adjustments, and status changes, so that regulatory compliance is strictly maintained.	**High**

Sprint Backlog – 2nd Sprint

1. Automated Batch Quarantine & Expiry Lock

Task ID	Task	Est. Hours
T - 01.1	Design quarantine batch management view on inventory dashboard	3
T - 01.2	Develop automated nightly cron job to detect and lock expired/near-expiry batches	4
T - 01.3	Implement hard database constraints preventing quarantined stock from cart checkout	3
T - 01.4	Add audit logging for disposal or return of quarantined pharmaceutical batches	2

T - 01.5	Test cron scheduler and expired batch quarantine enforcement	2

2. Barcode Scanning & Item Traceability

Task ID	Task	Est. Hours
T - 02.1	Design mobile/web barcode scanning interface for order packing stations	4
T - 02.2	Implement barcode decoding API linking scanned serials with active order items	4
T - 02.3	Build validation logic alerting staff on mismatched medication or batch scans	3
T - 02.4	Store verified serial tracking numbers against completed customer order records	2
T - 02.5	Test barcode scanner integration and pack validation flow	2

3. Cold-Chain Packaging & Protocol Tagging

Task ID	Task	Est. Hours
T - 03.1	Design cold-chain flag indicators and storage temperature warnings on product catalog	3
T - 03.2	Develop backend logic enforcing mandatory ice pack/insulated box checklists on packing	4
T - 03.3	Implement cold-chain manifest generator for logistics couriers	3
T - 03.4	Add temperature-sensitive handling advisory on customer digital receipts	2
T - 03.5	Test cold-chain packaging validation and courier manifest output	2

4. Secure Handover & Delivery OTP Verification

Task ID	Task	Est. Hours
T - 04.1	Design mobile OTP entry and customer signature pad interface for courier app	4
T - 04.2	Develop secure OTP generation and hashing API upon dispatch	3
T - 04.3	Implement OTP validation service with 3-attempt lockout protection	3

T - 04.4	Capture proof-of-delivery photo and geo-coordinates upon successful handover	3
T - 04.5	Test OTP validation, failure lockouts, and successful delivery completion flow	2

5. Geofenced Delivery Routing & Dispatch

Task ID	Task	Est. Hours
T - 05.1	Design courier dispatch dashboard with integrated Google/Mapbox maps	4
T - 05.2	Implement backend geofencing validation to verify delivery addresses within active zones	4
T - 05.3	Develop multi-stop delivery route optimization algorithm for courier batches	4
T - 05.4	Add real-time courier GPS coordinate telemetry endpoint	3
T - 05.5	Test geofence boundary checks and route calculation accuracy	2

6. Immutable Audit Trail & Regulatory Logs

Task ID	Task	Est. Hours
T - 06.1	Design administrator audit inspection table with multi-criteria filtering	4
T - 06.2	Implement AOP (Aspect-Oriented Programming) audit interceptors in Spring Boot backend	4
T - 06.3	Create tamper-proof append-only audit database schema for clinical actions	3
T - 06.4	Implement CSV/PDF export generator for NMRA regulatory compliance audits	3
T - 06.5	Test audit interceptor performance and verify log immutability	2

3rd Sprint – Week 11 to 12

Sprint ID	User Story	Priority
SP3 - 01	As a Customer, I want to set up automated monthly refill schedules for chronic illness medications (e.g., diabetes, hypertension), so that I never run out of vital daily prescriptions.	Medium
SP3 - 02	As a Pharmacist, I want to review and clinically renew pending chronic refill requests before automatic order dispatch, so that long-term patient medication safety is guaranteed.	Medium
SP3 - 03	As a Delivery Courier, I want to log cold-chain carrier box temperatures at dispatch and handover, so that temperature compliance records are maintained throughout transit.	Medium
SP3 - 04	As a Delivery Coordinator, I want to monitor active delivery fleets and receive automated delay or breach alerts in real time, so that logistics exceptions can be resolved immediately.	Medium
SP3 - 05	As a Pharmacy Assistant, I want to receive automated alerts when stock levels fall below critical reorder thresholds, so that inventory replenishments can be ordered proactively.	Medium
SP3 - 06	As a Finance Manager, I want to process authorized order cancellations and issue partial refunds through integrated payment pathways, so that customer disputes are resolved fairly.	Medium

Sprint Backlog – 3rd Sprint

1. Chronic Care Refill & Subscription Engine

Task ID	Task	Est. Hours
T - 01.1	Design subscription management UI (frequency, dosage, preferred delivery window)	4
T - 01.2	Develop automated recurring cron job scheduler for chronic refill order generation	4
T - 01.3	Implement payment pre-authorization and recurring billing gateway linkage	3
T - 01.4	Add customer subscription pause, skip, and cancel self-service controls	2
T - 01.5	Test recurring schedule accuracy and automated order creation workflow	2

2. Pharmacist Refill Clinical Renewal

Task ID	Task	Est. Hours
T - 02.1	Design chronic subscription clinical review queue for pharmacists	3
T - 02.2	Develop API to verify prescription validity and remaining allowable refills	3
T - 02.3	Implement doctor renewal notice trigger when prescription reaches maximum refill limit	3
T - 02.4	Log clinical refill approval timestamps and pharmacist digital signatures	2
T - 02.5	Test clinical refill approval and expiry warning workflows	2

3. In-Transit Cold-Chain Temperature Logging

Task ID	Task	Est. Hours
T - 03.1	Design temperature log submission form on courier mobile interface	3
T - 03.2	Develop API to record carrier cooler temperature readings with timestamps	3
T - 03.3	Implement automated breach threshold detection (>8°C or <2°C alert trigger)	3
T - 03.4	Link temperature logs with customer proof-of-delivery report	2
T - 03.5	Test temperature excursion breach alerts and logging integrity	2

4. Fleet Monitoring & Delivery Exception Alerts

Task ID	Task	Est. Hours
T - 04.1	Design live delivery coordinator dashboard displaying active courier locations	4
T - 04.2	Develop delivery delay estimation service based on geofence traffic anomalies	3
T - 04.3	Implement incident reporting modal for failed attempts, customer unavailable, or damaged goods	3
T - 04.4	Send automated customer notification updates during unexpected transit delays	2
T - 04.5	Test delivery exception workflow and coordinator alert dispatch	2

5. Low Stock Reorder Triggers & Alerts

Task ID	Task	Est. Hours
T - 04.1	Design reorder threshold configuration UI per medicine SKU	3
T - 04.2	Develop backend event listener triggering low stock flags upon order deduction	3
T - 04.3	Build automated purchase requisition draft generator for procurement team	3
T - 04.4	Send real-time low stock summary alerts to inventory managers via email	2
T - 04.5	Test threshold trigger precision and automated draft creation	2

6. Order Cancellation & Partial Refund Management

Task ID	Task	Est. Hours
T - 04.1	Design customer cancellation request interface with terms validation	3
T - 04.2	Develop refund calculation API handling full, partial, and restocking deductions	4
T - 04.3	Integrate payment gateway refund API endpoints (Stripe / Bank Gateway)	3
T - 04.4	Automatically restock unfulfilled batches back into active inventory pool	2
T - 04.5	Test cancellation edge cases, stock restorations, and payment refund webhooks	2

4th Sprint – Week 13 to 14

Sprint ID	User Story	Priority
SP4 - 01	As a Customer, I want to receive an OTP via SMS and email when my package is dispatched, so that I can securely verify the handover with the courier upon arrival.	Low
SP4 - 02	As a Operations Manager, I want to view operational dashboards showing prescription approval times and courier transit durations, so that workflow bottlenecks can be eliminated.	Low
SP4 - 03	As a Finance Manager, I want to track revenue, payment reconciliation, and pending COD collections, so that accurate daily financial audits are maintained.	Low
SP4 - 04	As a Operations Manager, I want to generate stock turnover and expiry wastage reports across all pharmaceutical categories, so that procurement efficiency is maximized.	Low
SP4 - 05	As a System Administrator, I want to monitor multi-channel notification gateway health (Email, SMS, WhatsApp), so that transactional messaging runs without interruption.	Low
SP4 - 06	As a System Administrator, I want to manage system backup routines and geofencing delivery boundaries, so that operational continuity and service coverage are maintained.	Low

Sprint Backlog – 4th Sprint

1. Multi-Channel Dispatch OTP Messaging

Task ID	Task	Est. Hours
T - 01.1	Design responsive transactional email and SMS notification templates for OTP delivery	4
T - 01.2	Develop asynchronous message producer service using RabbitMQ/Kafka queue	4
T - 01.3	Integrate SMS Gateway (Twilio/Dialog) and SMTP Email service providers	3
T - 01.4	Implement fallback gateway routing if primary SMS provider encounters failure	2

T - 01.5	Test multi-channel notification deliverability and message latency	2

2. Operational Performance Analytics Dashboard

Task ID	Task	Est. Hours
T - 02.1	Design operations management dashboard with interactive metric cards and charts	4
T - 02.2	Develop API aggregating prescription verification turnaround time (TAT) metrics	3
T - 02.3	Add courier fulfillment and average delivery duration tracking visualizations	3
T - 02.4	Implement date-range and regional delivery performance filters	2
T - 02.5	Test analytics aggregation accuracy against raw order database records	2

3. Financial Reconciliation & COD Auditing

Task ID	Task	Est. Hours
T - 03.1	Design financial ledger view displaying settled card payments vs pending COD	4
T - 03.2	Develop automated reconciliation API comparing courier collected cash with orders	3
T - 03.3	Implement COD collection handover sign-off and finance confirmation workflow	3
T - 03.4	Add financial summary export utility (Excel / PDF format)	2
T - 03.5	Test COD reconciliation balancing and financial audit report generator	2

4. Stock Turnover & Expiry Wastage Reporting

Task ID	Task	Est. Hours
T - 04.1	Design inventory intelligence dashboard layout with wastage loss charts	4
T - 04.2	Develop API calculating inventory turnover ratios by pharmaceutical therapeutic class	3
T - 04.3	Generate batch expiry projection and expired write-off valuation reports	2
T - 04.4	Add automated monthly inventory analytics email digest for executive management	2
T - 04.5	Test reporting query optimization on large dataset benchmarks	2

5. Notification Gateway Health Monitor

Task ID	Task	Est. Hours
T - 05.1	Design notification service health status and queue monitoring UI	3
T - 05.2	Develop health-check heartbeat endpoints for third-party SMS/Email gateways	3
T - 05.3	Implement automated dead-letter queue (DLQ) retry mechanism for failed messages	2
T - 05.4	Configure administrator alert when external gateway latency exceeds 5 seconds	2
T - 05.5	Test dead-letter retry logic and simulated gateway outage recovery	2

6. System Disaster Recovery & Geofence Config

Task ID	Task	Est. Hours
T - 06.1	Design administrative map tool for polygon geofence boundary drawing and editing	4
T - 06.2	Implement backend API to save and update active delivery polygon zones	3
T - 06.3	Configure automated daily MySQL database snapshot and offsite backup routines	2
T - 06.4	Implement point-in-time database restoration runbook scripts	2

T - 06.5	Perform disaster recovery simulation test and geofence polygon boundary verification	2